MIL-C-25/15C
21 June 1968
SUPERSEDING
MIL-C-25/15B
29 September 1960

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PAPER-DIELECTRIC, DIRECT CURRENT (HERMETICALLY SEALED IN METAL CASES), STYLE CP91

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-25.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

REQUIREMENTS:

Dimensions:
Capacitors: See figure 1 and tables I and IV.
Brackets: See figure 2 and table III.

Case and bracket material:
Magnetic or nonmagnetic.
Capacitors shall be uninsulated.

Design and construction: Capacitors shall be of the design, construction and physical dimensions as specified in figure 1 and tables I and IV.

Capacitance value: See table IV.

Capacitance tolerance: See table IV.

DC voltage rating: See table IV.

Operating temperature range:
Characteristics E and F - -55° to +85° C.
Characteristic K - - - - - -55° to +125° C.

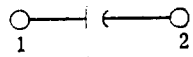
Vibration grade: 1 (10 to 55 Hz, incl.)

Terminal:

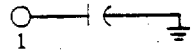
Symbol B - - - - Solder lug (nonremovable).

Circuit diagram:

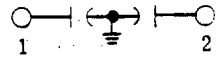
Symbol 1 - - - -



Symbol 2 - - - -



Symbol 6 - - - -



Voltage derating with temperature: See figure 3.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
Terminal to terminal - As specified in table V.
Terminal to case (when case is not a terminal) - As specified in table V.
Capacitors shall withstand specified voltages with no permanent damage or open- or short-circuiting.

Insulation resistance: Method 302 of MIL-STD-202.
At 25° C:
Terminal to terminal - As specified in table VI.
Terminal to case (when case is not a terminal) - Shall exceed 3,000 megohms.
At 85° C (characteristics E and F):
Terminal to terminal - As specified in table VI.
At +125° C (characteristic K):
Terminal to terminal - As specified in table VI.

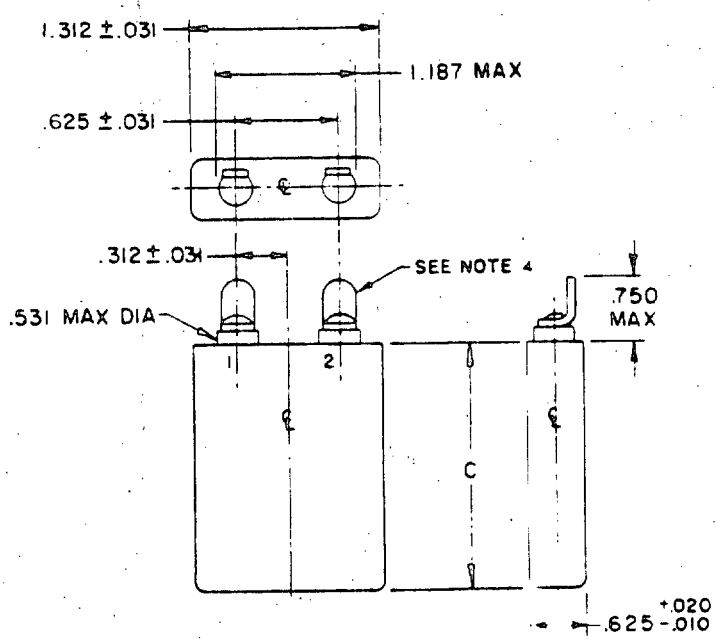
Capacitance: Method 305 of MIL-STD-202.
Measured capacitance shall be within the tolerance specified in table IV.

Dissipation factor: When measured as specified in MIL-C-25, shall not exceed 1 percent.

Barometric pressure (flashover):
Method 105 of MIL-STD-202.
Test condition - B (50,000 ft.)
Test potential - 125 percent of rated dc voltage applied for a minimum of 1 minute without visible damage, external flashover, or open- or short-circuiting.

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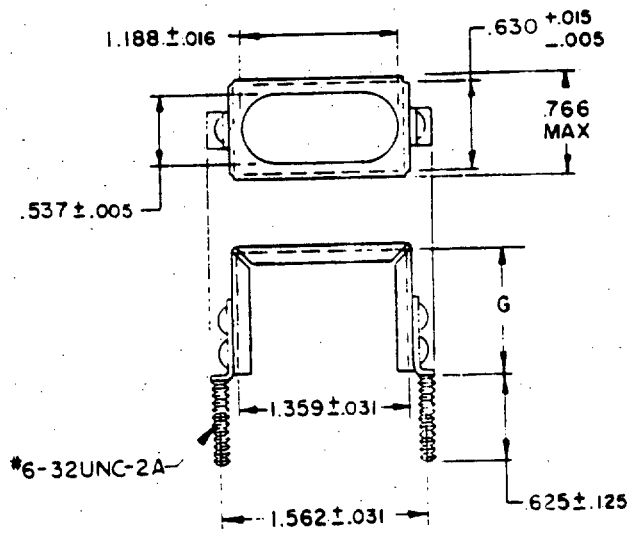


IN	MM
.010	.25
.020	.51
.031	.79
.312	7.92
.531	13.49
.625	15.88
.750	19.05
1.187	30.18
1.312	33.32

NOTES:
1. See table I for additional dimensions.
2. All dimensions in inches.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Terminals need not be identified by numbers as shown. Shape of terminals and terminal lugs (drilled, punched, forked, or slotted) optional.
5. For circuit 2, use terminal 1.

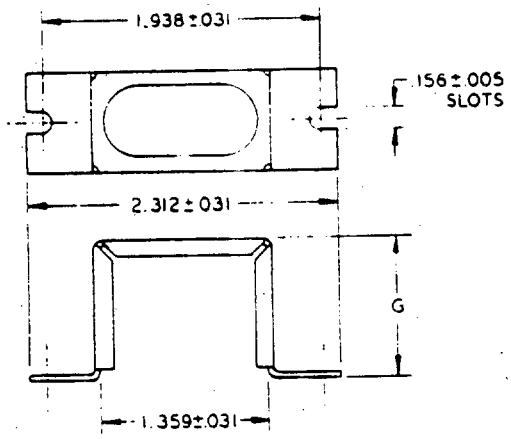
FIGURE 1. Style CP91 capacitor.

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SPADE-LUG BRACKET

INCHES	MM
.005	.13
.015	.38
.016	.41
.031	.79
.125	3.18
.156	3.96
.537	13.64
.625	15.88
.630	16.00
.766	19.46
1.188	30.18
1.359	34.52
1.562	39.67
1.938	49.23
2.312	58.72

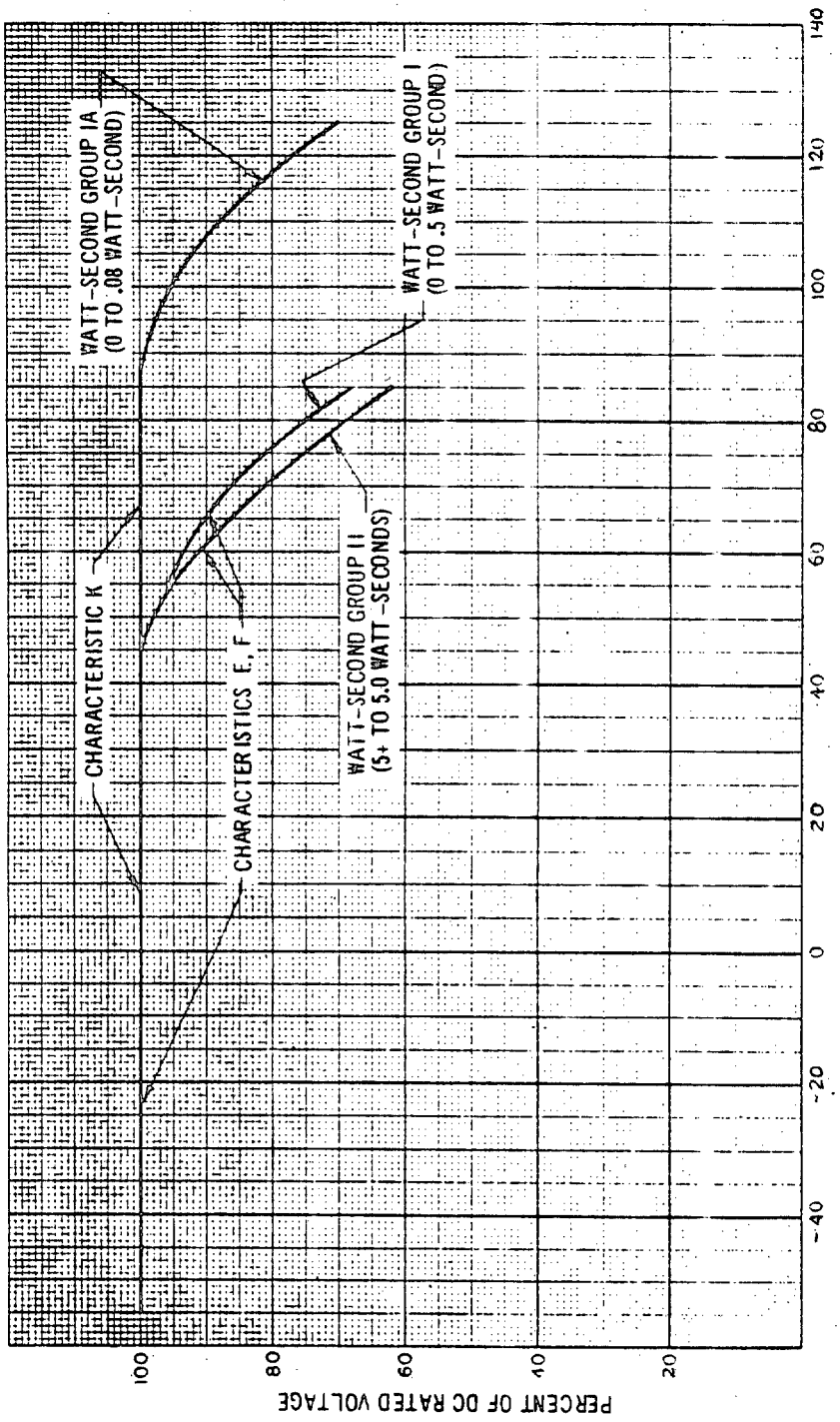


FOOTED BRACKET

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
3. Shape of bracket optional provided specified dimensions are met. Spade lugs shall be rigid and double-welded, double riveted, single riveted and soldered, or welded and single riveted to the bracket.

FIGURE 2. Style CP091 brackets.

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Note: Permissible operating voltages, while based on incomplete data, are the capacitor suppliers' best estimate to provide a life expectancy of 8,800 hours of continuous operation at higher ambient temperatures. Longer life can be expected by operation at voltage lower than indicated on the curve; for example, a life expectancy of 44,000 hours may be obtained by operation at 70 percent of the voltage determined by use of this curve. Also, a life longer than 8,800 hours may be expected at the voltage determined by use of this curve if the high ambient temperature prevails for only a portion of the whole operating time. No derating is necessary for capacitors of characteristic K for 2,000 hours of operation at 125°C.

FIGURE 3. Voltage derating for ambient temperature (characteristic E, F, and K)

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TABLE I. Style CP91 case dimensions and bracket designations.

Case size	Dimensions L ± 0.031	Bracket designation 1/ (see fig. 3)
B1	1.062	CP091-B1
B2	1.375	CP091-B2
B3	1.625	CP091-B3
B4	2.000	CP091-B4
B5	2.500	CP091-B5
B6	2.750	CP091-B6

1/ Complete bracket designation shall include additional symbol to indicate design of bracket.

TABLE II. Millimeter equivalents of decimal inches.

In.	mm	In.	mm
0.031	0.79	1.892	48.06
0.954	24.23	2.000	50.80
1.062	26.97	2.392	60.76
1.267	32.18	2.500	63.50
1.375	34.93	2.642	67.11
1.517	38.53	2.750	69.85
1.625	41.28		

TABLE III. Style CP091 bracket dimensions.

Bracket designation 1/2/	Bracket dimension G ± 0.031	Case size
CP091-B1	0.954	B1
CP091-B2	1.267	B2
CP091-B3	1.517	B3
CP091-B4	1.892	B4
CP091-B5	2.392	B5
CP091-B6	2.642	B6

1/ Bracket shall be removable.

2/ Complete bracket designation shall include additional symbol to indicate design of bracket.

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TABLE IV. Style CP91 characteristics and case size designations.

Type designation [1]	DC voltage rating (see fig. 3)	Capacitance	Capacitance tolerance [2]	Characteristics	Vibration grade	Case size		
						Characteristic E	Characteristic F	Characteristic K
	Volts	µf						
CP91B-KE104K1	400	0.1	K	K	1	---	---	B1
CP91B-KE254K1	400	0.25	K	K	1	---	---	B2
CP91B1KE504K1	400	0.5	K	K	1	---	---	B4
CP91B-E105K1	400	1	K	E, F, K	1	B5	B4	B6
CP91B-EF503K1	600	0.05	K	E	1	B1	---	---
CP91B-EF254K1	600	0.25	K	E	1	B3	---	---
CP91B-EF504K1	600	0.5	K	E	1	B4	---	---
CP91B-F105K1	600	1	K	E, F	1	B6	B5	---
CP91B-EG103K1	1,000	0.01	K	E	1	B1	---	---
CP91B-EG203K1	1,000	0.02	K	E	1	B1	---	---
CP91B-EG503K1	1,000	0.05	K	E	1	B2	---	---
CP91B-EG104K1	1,000	0.1	K	E	1	B2	---	---
CP91B-EG254K1	1,000	0.25	K	E	1	B4	---	---
CP91B-G504K1	1,000	0.5	K	E, F	1	B6	B5	---
CP91B6EE503V1	400	0.05-0.05 [3]	V	E	1	B1	---	---
CP91B6EE104V1	400	0.1 -0.1	V	E	1	B2	---	---
CP91B6EE504V1	400	0.5 -0.5	V	E	1	B5	---	---
CP91B6EF503V1	600	0.05-0.05	V	E	1	B2	---	---
CP91B6-F104V1	600	0.1 -0.1	V	E, F	1	B3	B2	---
CP91B6EF254V1	600	0.25-0.25	V	E	1	B4	---	---
CP91B6-F504V1	600	0.5 -0.5	V	E, F	1	B6	B5	---
CP91B6EG103V1	1,000	0.01-0.01	V	E	1	B1	---	---
CP91B6EG203V1	1,000	0.02-0.02	V	E	1	B2	---	---
CP91B6EG503V1	1,000	0.05-0.05	V	E	1	B2	---	---
CP91B6-G104V1	1,000	0.1 -0.1	V	E, F	1	B4	B3	---
CP91B6-G254V1	1,000	0.25-0.25	V	E, F	1	B6	B5	---
CP91B6EH103V1	1,500 [4]	0.01-0.01	V	E	1	B1	---	---
CP91B6EH503V1	1,500	0.05-0.05	V	E	1	B4	---	---

[1] Complete type designation shall include additional symbols to complete style designation and to indicate circuit (where applicable) and characteristic (where applicable).

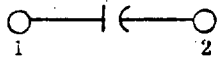
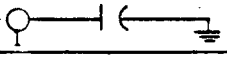
[2] Capacitance tolerance in percent: K, ±10; V, +20, -10.

[3] For multiple-unit capacitors, the watt-second rating is the sum of the watt-second ratings of the component sections.

[4] For derating purposes, watt-second group I is limited to capacitors having dc voltage rating of 1,000 volts or less. Watt-second group II includes capacitors having a watt-second rating of 0.5 and less which are not included in watt-second group I.

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TABLE V. Dielectric withstanding voltage test.

Circuit symbol	Circuit diagram	Test	Test connections	Test voltage (percent of rated dc voltage)	Time test applied (minutes)
1		Terminal to terminal	1 to 2	200 1/	1 2/
		Terminal to case	1 and 2 to case	400 5/	1 3/ 4/
				200 + 1,000 6/ volts	1 3/ 4/
2		Terminal to terminal	1 to case	200	1 2/
6		Terminal to terminal	1 and 2 to case	200	1 7/
			1 to 2	200	1 7/

1/ 175 percent of rated dc voltage after temperature-and immersion-cycling and moisture resistance tests.

2/ For quality conformance inspection and at the option of the supplier, either 200 percent of rated dc voltage shall be applied for 15 seconds or 250 percent of rated dc voltage for not less than 1 second.

3/ For quality conformance inspection, the period of test voltage application shall be 1 second.

4/ For quality conformance inspection, applications may be made between each terminal individually and the case, at the option of the supplier.

5/ For capacitors having dc voltage ratings of 600 volts and less.

6/ For capacitors having dc voltage ratings greater than 600 volts.

7/ For quality conformance inspection, the application may be made for 1 second if the capacitor has passed group A inspection.

TABLE VI. Terminal to terminal insulation resistance measurements.

Capacitance values	Minimum insulation resistance 1/		
	Characteristic E	Characteristic F	Characteristic K
	At 25° C	At 25° C	At 25° K
0.33 microfarads and less	6,000 megohms	4,500 megohms	18,000 megohms
greater than 0.33 microfarads	2,000 megohm-microfarads 2/	1,500 megohm-microfarads 2/	6,000 megohm-microfarads 2/
	At 85° C	At 85° C	
0.033 microfarads and less	600 megohms	---	---
greater than 0.033 microfarads	20 megohm-microfarads 2/	15 megohm-microfarads 2/	---
	At 125° C	At 125° C	At 125° C
greater than 0.067 microfarads	---	---	10 megohm-microfarads 2/

1/ A potential equal to the dc rated voltage or 500 volts dc, whichever is less, shall be applied.

2/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

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Vibration (grade 1, 10 to 55 Hz incl.):
Method 201 of MIL-STD-202.
Duration - 3 hours (60 minutes in each direction).
Measurements - An electrical signal of 1±0.2 kHz, at a level of 1±0.5 volts applied during the last 30 minutes in each direction. There shall be no momentary arcing or breakdowns, open- or short-circuiting, or mechanical damage.

Salt spray (corrosion): Method 101 of MIL-STD-202, test condition B (48 hours), with no evidence of corrosion or obliteration of marking.

Temperature cycling: Method 102 of MIL-STD-202, except that the test temperature of step 3 shall be as follows:
Characteristics E and F: +85° C.
Characteristic K: +125° C.

Immersion: Method 104 of MIL-STD-202.
Test condition C, except that the duration of each immersion shall be 30 minutes, and change between solutions shall be made in not more than 3 seconds.
Measurements after final cycle:
Dielectric withstanding voltage, at 25° C:
Terminal to terminal - As specified in table V.
Terminal to case (when case is not a terminal) - As specified in table V.
Insulation resistance, at 25° C:
Terminal to terminal - Not less than 30 percent of the value specified in table VI.
Terminal to case (when the case is not a terminal) - Not less than 1,500 megohms.

Terminal strength: Method 211 of MIL-STD-202, test conditions A and E.
Test condition A (pull): Applied force - 5 pounds.
Test condition E (torque) - 2 pound -inches.

Moisture resistance: Method 106 of MIL-STD-202.
Polarization voltage - 100 volts.
Final measurements:
Dielectric withstanding voltage and insulation resistance requirements - As specified in immersion.

Seal:
Method 112 of MIL-STD-202, test condition A, except that bath temperature shall be as follows:
For characteristics E and F: -85±3° C.
For characteristic K: +125±3° C.
Liquid-filled capacitors - 10 minutes on each of two opposite sides at test temperatures as follows:
Characteristics E and F: -85°±3° C.
Characteristic K: +125°±3° C.
There shall be no evidence of leakage.

Low temperature and capacitance change with temperature:
Low temperature -
Conditions: -55° ± 3° C for 48 ± 4 hours with rated dc voltage applied.
Capacitance change with temperature:
Characteristic E - Not to exceed ± 10 percent.
Characteristic F - Not to exceed ± 10, -30 percent.
Characteristic K - Not to exceed +75, -10 percent.

Life:
Method 108 of MIL-STD-202, test condition B (250 $^{+8}_{-0}$ hours).
Electrical load: 140 percent of rated dc voltage.
Final measurements (at 25° C):
Insulation resistance:
Terminal to terminal - Not less than 30 percent of the value specified in table VI.
Capacitance:
Characteristics E and F - Within 10 percent of initial value.
Characteristic K - Within 5 percent of initial value.
Dissipation factor:
Characteristics E and K - Not to exceed 1 percent.
Characteristic F - Not to exceed 1.5 percent.

Inspection lot: An inspection lot may consist of capacitors of the same characteristics described herein, and the same characteristics of styles CP67 and CP69.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 11

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